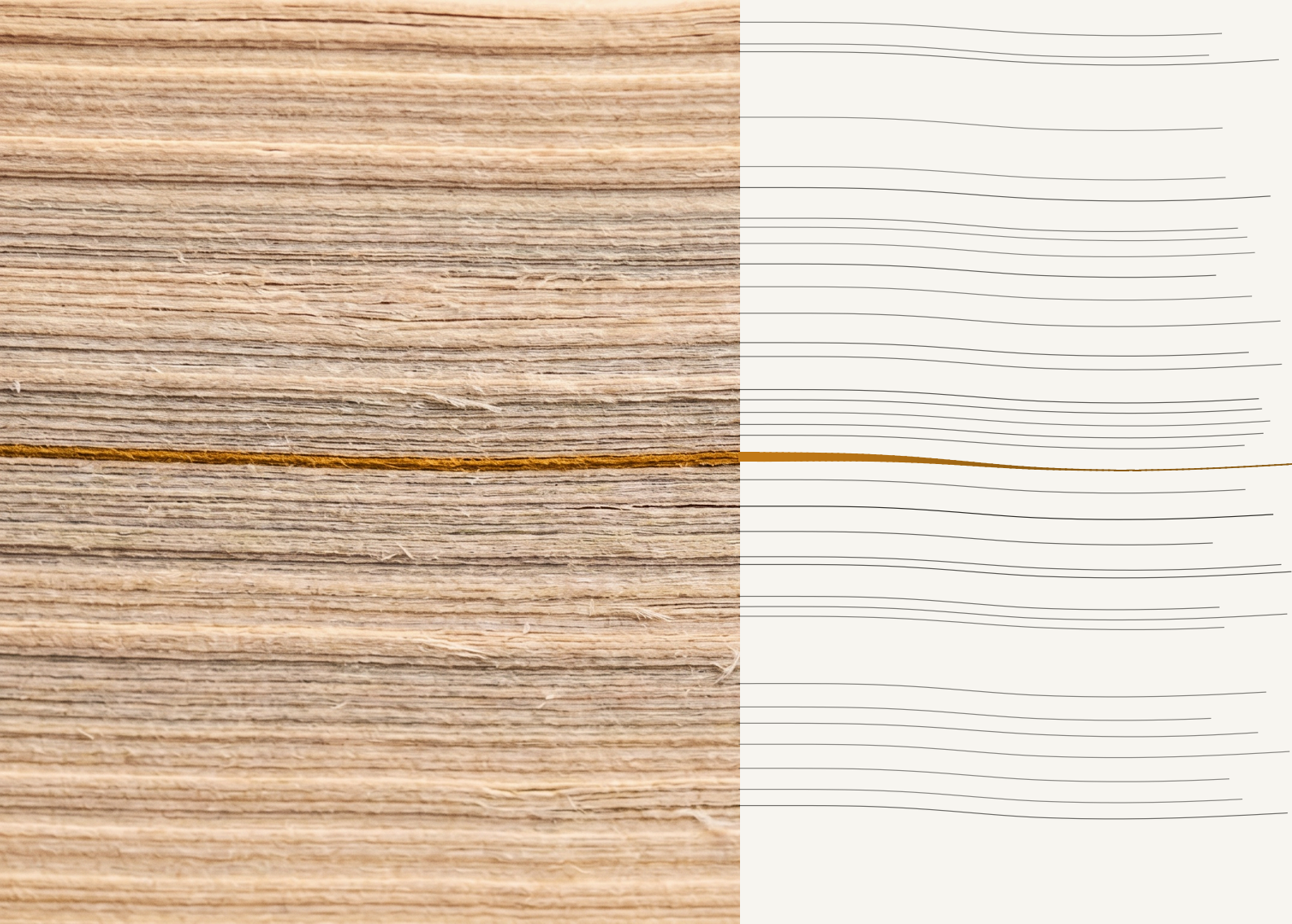




J | JULIE

An auditable, LLM-assisted workflow for systematic reviews

The guiding principle is that the whole process should be reproducible and auditable.

Why this system exists

Systematic reviews are slow and expensive at exactly the points where methodological rigor matters most: dual screening of records, data extraction, and appraisal of the evidence. Large language models can accelerate these steps, but acceleration is only defensible to peer reviewers and guideline panels if three conditions hold. The method must be standard, every step must be traceable, and the report must state precisely what a machine checked, what a person decided, and what nobody verified.¹

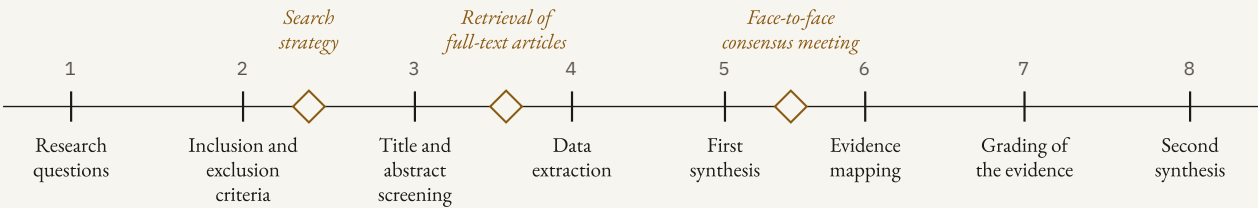
¹ The methods report can therefore separate machine checks from human decisions.

The system described here is a research assistant that accompanies a review team from the research question to a draft manuscript under these conditions. It does not replace the reviewers. It reproduces the structure of a conventional systematic review, automates the repetitive parts, and stops wherever a human judgment is required.

The guiding principle is that the whole process should be reproducible and auditable. Calculations can be recomputed from preserved inputs, rules and software environment. For every language-model step, the input, the recorded output and the available model identifiers remain inspectable, and the limits of exact re-execution are stated. The depth of human audit is a documented choice of the research team, not a setting imposed by the software; the system's task is to inform that choice with evidence and to report it honestly.

The eight steps and the human gates

The system models the actual steps of a systematic review. The human hand-over points are the interfaces of the chain: automation stops there and resumes only after a named person has acted.



#	STEP	WHO DOES IT
1	Research questions (PICO-structured)	Human
2	Inclusion and exclusion criteria	Machine drafts, human approves
◇ GATE	<i>Search strategy: planning, final approval, database searches, export hand-over</i>	Assistant helps plan; human approves and runs the searches
3	Title and abstract screening	Two language models, human adjudication
◇ GATE	<i>Retrieval of full-text articles</i>	Human
4	Data extraction (evidence items)	Machine, guided by the research questions
5	First synthesis: draft recommendations and a brief for the consensus meeting	Machine
◇ GATE	<i>Face-to-face consensus meeting</i>	Human panel
6	Evidence mapping (supporting and contradicting evidence per recommendation)	Machine
7	Grading of the evidence	Machine, deterministic rules
8	Second synthesis: manuscript draft	Machine; read and approved by humans

Beyond these, further named human decisions sit inside the chain: the reference standard for screening performance, the written record of the consensus meeting, the choice of model when a provider declines a request, and the final reading of the manuscript. The accurate description of the system is therefore not “it runs end to end”, but:

the machine stages run through, and at every human gate a named person decides, and that decision leaves a record.

03

Two alternative grading schools

Step 7 can follow one of two established frameworks. The choice is made per project at the outset, because it determines the extraction schema, the grading rules and the output; it is then frozen for every subsequent run.

	OXFORD CEBM	GRADE
<i>Unit of analysis</i>	The individual article	The body of evidence (question × outcome)
<i>Output</i>	Level of evidence and strength of recommendation	Certainty: high, moderate, low, very low
<i>Typical context</i>	Guideline development with task force voting	Evidence summaries and evidence-to-decision frameworks
<i>Risk of bias</i>	Reported separately	Part of the certainty rating, using established instruments

A project may also apply both schools to the same corpus. In that case each school produces its own workbook, its own methods report and its own PRISMA flow diagram, and the manuscript cites one of them by explicit choice.

Design principles

01

The language model states facts; deterministic code judges.²

The model answers factual questions about each article and supports every answer with a verbatim quotation from the source. Levels of evidence, risk-of-bias judgments and certainty ratings are computed by tested, deterministic rules from those extracted facts. When two reviewers disagree, it is because the facts they extracted differ, never because the rules did. Title and abstract screening is the one step where the model output is itself the decision; there, the machine's role is to compare two independent decisions and route disagreements to a person.

² Every result can be traced back to its verbatim source quotation, the rule that produced it, the configuration in force and the record of the run.

02

Two independent reviewers, with human adjudication.

Screening is performed by large language models from two independent providers, mirroring the dual-reviewer standard. Their agreement is measured, not assumed. Where they disagree or are uncertain, a human decides, by default in a blinded view that does not show what the models concluded.

Their agreement is measured, not assumed.

03

Every human decision carries a name.

Approvals, adjudications, consensus outcomes and deliberate deviations are recorded with the name of the person who made them. Machine or placeholder names are refused. The methods report can therefore separate machine checks from human decisions.

Machine or placeholder names are refused.

04

Absence is not exclusion.

The system keeps three situations apart: “no evidence was found” (a finding), “an error occurred” (not a finding; the step must be repeated), and “not yet processed” (work outstanding). An article that could not be read is recorded as a named gap, not silently excluded. A document with unresolved gaps is not released as if it were complete.

A document with unresolved gaps is not released as if it were complete.

05

A full audit trail.

Every result can be traced back to its verbatim source quotation, the rule that produced it, the configuration in force and the record of the run. Generated files carry a fingerprint of their inputs, and the system refuses to build on an input that has changed since.

...the system refuses to build on an input that has changed since.

06

Paid runs are gated.

No step that calls a commercial model starts without a cost estimate, a spending ceiling and an approval given by a named person. A refusal to spend is a normal outcome, not an error.

A refusal to spend is a normal outcome, not an error.

07

Methodological deviations are declared.

Any departure from the standard method is itself a recorded configuration decision and appears in the methods section.

05

The researcher's experience

Researchers can work through the process in two equivalent ways: by issuing the steps directly, or through a conversational assistant that reads the state of the project, explains in plain language what comes next, runs the free steps, prepares the paid ones, and records the researcher's stated decisions. The assistant leads; the human decides; the project folder remembers. Bulk tasks such as adjudicating hundreds of screening conflicts remain in spreadsheets, where they are faster and less prone to anchoring.

*The assistant leads; the human decides;
the project folder remembers.*

What the system does not do

It does not run database searches. The assistant helps design search strategies from the research question; the final approval and the execution of the searches are the researchers' task. The reporting obligation remains: the full search strategy for each database (PRISMA 2020 items 6 and 7) belongs in the manuscript, and the methods report states that it expects this from the authors.

It does not decide recommendations. Every grade and every draft is a labeled proposal. The final wording and strength of each recommendation are voted on by the panel.

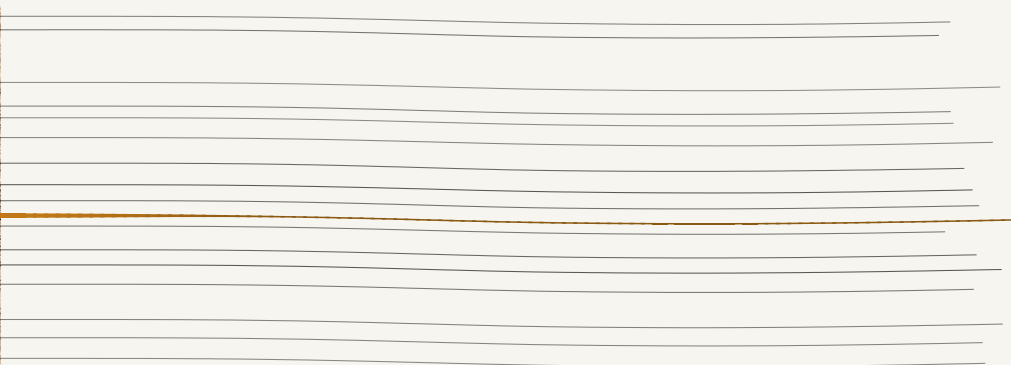
It does not supply clinical meaning. Automated checks verify citations, numbers, grading vocabulary and quotation leakage. Whether a sentence is medically correct, or accurate but misleading, is for the clinical reader to judge.

It does not replace expert reading. A passing automated check means the system is internally consistent, not that the output is right.

Current status

The complete chain, from search planning to manuscript draft, has been verified offline for both grading schools using synthetic inputs, scripted model responses and scripted human decisions, and has been exercised on a calibration corpus from an earlier consensus project. This establishes that the software behaves as specified; it is not yet a validation of output quality on a real review, which is the next stage of work.³

³ A passing automated check means the system is internally consistent, not that the output is right.



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